

JUG BAY, MD, USA

WMO: 997994

Lat: 38.781N

Lon: 76.708W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.1	18.7	-2.3	4.9	17.2	4.3	6.9	24.5	16.8	40.6	15.7	36.6	4.5	310	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	93.4	77.9	90.9	76.7	88.6	75.4	80.0	89.9	78.5	87.9	77.3	86.2	3.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	142.2	84.6	76.0	136.0	83.2	74.8	130.4	82.0	43.4	89.9	42.0	87.9	40.7	86.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.7	12.5	10.9	DB	9.8	98.6	5.1	4.0	6.1	101.5	3.1	103.8	0.2	106.1	-3.5	109.0	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.0	33.8	38.1	44.5	56.7	73.9	78.2	75.7	70.2	58.1	46.6	41.3
		DBStd	16.91	9.25	8.50	9.20	8.46	7.67	5.54	4.18	6.29	7.59	7.90	8.83
		HDD50	1580	507	348	224	26	1	0	0	0	18	157	299
		HDD65	4269	967	754	638	277	84	4	0	22	236	554	733
		CDD50	4150	5	14	53	227	503	717	875	797	605	270	54
		CDD65	1364	0	0	1	29	121	270	410	332	177	23	1
		CDH74	11869	0	0	21	335	994	2412	3904	2695	1329	174	5
		CDH80	4552	0	0	2	107	271	927	1726	1031	451	37	0
(14)	Wind	WSAvg	3.7	4.8	5.2	5.1	4.6	3.7	2.8	2.4	2.1	2.4	3.1	4.0
(15)	Precipitation	PrecAvg	44.6	3.1	2.7	3.9	3.5	3.9	4.4	4.3	4.1	3.5	2.8	3.6
(16)		PrecMax	63.3	6.5	7.0	9.4	5.6	10.4	10.2	7.4	10.5	10.8	8.1	6.5
(17)		PrecMin	33.3	1.3	0.5	0.0	1.9	1.1	2.0	1.8	1.4	0.0	0.2	0.4
(18)		PrecStd	6.2	1.2	1.5	1.8	1.1	2.0	2.2	1.6	1.9	2.8	2.0	1.7
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	65.6	70.4	76.7	89.6	90.0	95.5	98.4	94.4	92.5	84.5	73.9	69.7
(20)		0.4% MCWB	55.4	58.0	61.8	65.5	75.3	79.1	79.1	76.8	76.1	70.5	62.2	64.2
(21)		2% DB	58.9	62.8	70.7	82.7	86.1	91.4	94.4	91.1	89.2	78.7	67.5	65.4
(22)		2% MCWB	51.7	52.6	59.5	66.3	73.2	77.2	78.6	76.3	75.1	66.4	58.3	60.3
(23)		5% DB	52.6	57.0	65.1	77.3	83.2	88.6	91.3	88.5	85.5	74.5	63.2	59.9
(24)		5% MCWB	46.2	48.1	56.0	63.9	70.5	75.5	77.5	74.9	73.6	65.0	56.9	55.8
(25)		10% DB	47.8	51.4	59.6	72.5	80.2	85.7	88.7	86.2	81.8	70.9	59.4	55.3
(26)		10% MCWB	42.8	43.9	51.0	60.5	69.0	73.9	76.3	73.6	71.1	63.3	53.7	51.3
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	59.9	60.5	64.3	70.9	77.1	81.0	83.2	80.4	78.4	73.0	65.1	66.1
(28)		0.4% MCDB	64.8	68.5	73.6	78.0	88.5	92.0	92.9	89.9	88.2	80.7	69.6	68.9
(29)		2% WB	52.7	55.4	60.6	67.9	75.0	79.0	80.7	78.1	76.9	70.4	61.3	60.6
(30)		2% MCDB	57.6	62.6	68.1	76.4	84.5	88.7	90.2	86.7	85.7	74.8	65.6	63.8
(31)		5% WB	47.8	50.0	57.3	64.4	73.1	77.4	79.3	76.9	75.5	68.5	58.4	57.0
(32)		5% MCDB	51.3	55.3	64.9	73.6	81.3	86.9	89.0	84.7	83.3	73.2	62.3	59.6
(33)		10% WB	43.2	46.0	52.8	61.6	71.1	75.8	78.0	75.4	74.0	65.0	54.9	51.4
(34)		10% MCDB	47.4	51.9	57.6	69.2	78.0	83.7	87.0	82.3	79.5	70.0	58.6	54.5
(35)	Mean Daily Temperature Range	MDBR	16.7	17.8	19.0	21.5	20.8	20.7	20.4	20.3	20.3	19.6	19.0	16.4
(36)		5% DB MCDBR	23.7	27.1	27.6	27.4	23.3	23.8	23.5	22.7	23.9	24.6	24.0	20.9
(37)		5% DB MCWBR	18.1	17.3	17.1	12.4	11.2	10.7	9.5	9.1	9.8	12.7	15.9	16.5
(38)		5% WB MCDBR	23.0	24.6	25.3	22.0	21.1	21.6	21.9	19.5	19.6	19.4	20.8	19.5
(39)		5% WB MCWBR	19.6	17.8	17.5	12.9	10.4	10.2	9.5	8.9	9.2	12.3	15.9	16.5
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.321	0.356	0.397	0.438	0.492	0.517	0.509	0.420	0.361	0.325	0.310
(41)		taud	2.495	2.449	2.407	2.310	2.226	2.121	2.067	2.075	2.297	2.466	2.525	2.531
(42)		Ebn at Noon	276	287	286	279	269	253	246	244	261	269	268	267
(43)		Edh at Noon	25	30	34	40	44	49	52	50	38	29	24	23
(44)	All-Sky Solar Radiation	RadAvg	672	938	1249	1581	1787	1975	1935	1724	1401	1052	763	590
(45)		RadStd	53	103	97	118	206	150	95	95	156	115	65	55

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (77 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+234	+139

Nomenclature: See separate page